

Improved 1:1 / P2P Mobile Layout

Feature brief · VERA Web (vonage-video-react-app)

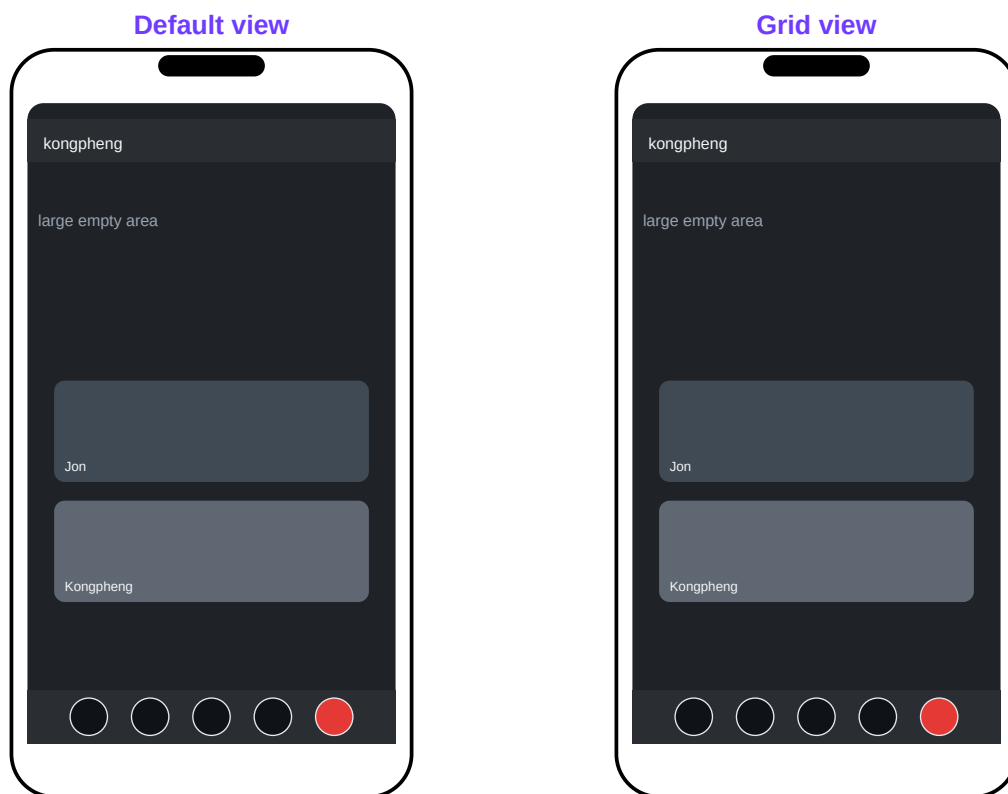
Background

Not every customer will ship a native mobile application. Many users rely on web applications to access video-meeting features from their phone, especially on devices that don't support or don't want another app. Optimising the VERA Web experience for mobile browsers therefore matters — regardless of the user's platform choice, the interface must feel native, efficient, and familiar.

Problem

Today's VERA layout manager works well for multi-party calls, but the 1:1 experience on mobile browsers is inconvenient. The local and remote tiles are rendered **side-by-side (or stacked)**, leaving large empty areas on screen and forcing both faces to compete for the same limited space. That is not the standard pattern for 1:1 calls, and it violates the expectations users bring from WhatsApp, FaceTime, Google Meet or Zoom.

Reference — current VERA on Chrome for iPhone (1:1):



Proposal

Introduce a **doctor-patient / student-teacher** layout dedicated to 1:1 calls on mobile: the remote participant fills the canvas edge-to-edge, the local self view becomes a small floating picture-in-picture tile. The layout turns itself on and off automatically; it is never user-selectable.

Current (side-by-side tiles)

Proposed (remote full-bleed + self PiP)



When it engages

- **Constrained device:** real mobile (UA-detected) **or** viewport ≤ 768 px
- **Exactly one remote participant** (strict 1:1)
- **No content / screen sharing** active
- **The remote isn't a screenshare stream**

Any breach — a third participant joins, a screenshare starts, or the desktop viewport grows past 768 px on a non-mobile device — auto-reverts to the existing grid / active-speaker layout. Nothing about the multi-party flow changes.

Self PiP behaviour

- **Default corner:** bottom-right, matching WhatsApp / Google Meet / Zoom. Resets each call (no persistence).
- **Drag & corner-snap:** the user can drag the self tile to any of the four corners; on release the tile animates to the nearest corner.
- **Toolbar-aware:** snap zones are clamped so the PiP never overlaps the bottom toolbar.
- **Source aspect ratio:** the PiP takes its aspect from the local camera's `videowidth / videoHeight` — the self-view never letterboxes.
- **Always visible:** the self PiP stays on screen for the entire 1:1 call, matching the native pattern across WhatsApp, FaceTime, Google Meet and Zoom.
- **Tap-to-reveal controls:** a single tap on the self PiP surfaces a camera-flip icon (bottom-left, mobile only) and a video-effects icon (bottom-right). Both auto-hide after 3 s idle.
- **Same visual language as the regular self-view tile:** audio-level meter top-right, mic-muted icon in the same slot, mirror transform, initials fallback when the camera is off. No name label —

space is too tight.

Why this matters

- Matches the 1:1 UX customers already know from native mobile apps — no retraining needed.
- Removes the empty-space awkwardness of today's mobile 1:1 view.
- Preserves full control for multi-party and screenshare scenarios — zero regression risk on those paths.
- Keeps the feature invisible when it shouldn't apply: desktop multi-party calls look exactly the same as before.

References

- WhatsApp (native iOS) — remote full-bleed, self PiP bottom-right
- FaceTime (native iOS) — remote full-bleed, self PiP draggable, auto-snaps
- Google Meet (Chrome on iPhone) — remote dominant, self PiP floating
- Zoom (web on mobile) — remote dominant, self PiP floating